

Lab Assignment 4
Machine Learning Lab - CSDC0336
(NITJ-CSED)

Learning Goals:

- Load and explore a recent real-world dataset using Pandas.
- Implement gradient descent from scratch for linear regression.
- Visualize learning process and effect of learning rate.
- Compare custom implementation with scikit-learn's Linear Regression.

Dataset: World Happiness Report 2023

The World Happiness Report ranks countries by happiness based on factors like GDP, social support, life expectancy, freedom, generosity, and perceptions of corruption. The data is from 2023 and is ideal for regression analysis.

Dataset Link: <https://www.kaggle.com/datasets/ajaypalsinghlo/world-happiness-report-2023>

Task 1: Load the dataset into a Pandas DataFrame. Compute summary statistics and check for missing values. Create a correlation matrix and identify which feature has the strongest linear correlation with Life Ladder. Generate a scatter plot of that feature vs happiness score. Create a histogram of the target variable. Split data into training (80%) and testing (20%) sets using `random_state=42`. Document your steps with markdown.

Task 2: Implement gradient descent from scratch using your chosen feature. Write functions to compute gradients and perform updates. Try learning rates: 0.001, 0.01, 0.1 with 100 iterations. Plot cost vs iterations for each rate and choose the best. Plot the final regression line over the scatter plot. Compare your model's parameters and test MSE with scikit-learn's Linear Regression. Write a brief conclusion explaining what your model reveals about the relationship between your chosen feature and national happiness.

Task 3: Create a new feature by combining two existing features that you think might together influence happiness (e.g., $\text{Log GDP per capita} * \text{Social support}$ or $\text{Freedom to make life choices} + \text{Positive affect}$). Explain why you chose this combination. Run gradient descent again using this new feature as the sole predictor. Compare its performance (test MSE, R^2) with the single-feature model from Task 2. Also, create a scatter plot of the new feature vs happiness and overlay the regression line. Discuss whether the engineered feature improved predictions and why.